Individual decisions and adding up

Advanced Macroeconomics

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Plan

- Plan: analyze a central decisions households make: consumption vs. saving
- Macroeconomic variables (consumption, investment) should be viewed as aggregation of the variables we analyze here.
- We will learn how to analyze these choices numerically.

A Two Period Model

The simplest two period model

Two dates $t = 0, 1$

Preferences:

- felicity $u(c)$ strictly increasing
- utility from consumption plan (c_0, c_1) with $c_t \geq 0$:

$$u(c_0) + \beta u(c_1)$$

Receive income y_0 date 0, y_1 at date 1

Save or borrow via one period bonds with interest rate R

- Save/borrow 1 unit of consumption at date 0, get/repay R units at date 1
- R^{-1} = price of bond that pays 1 unit of consumption next period. For now given (PE).

Budget constraints for date $t = 0, 1$

$$c_0 + R^{-1} a_1 = a_0 + y_0, \quad a_0 \text{ given}$$

$$c_1 = a_1 + y_1$$

Optimal consumption and bond choice

Solve:

$$\max_{c_0, c_1, b_0} \; u(c_0) + \beta u(c_1)$$

$$c_0 + R^{-1} a_1 = a_0 + y_0 := w_0 = \text{“cash on hand”}$$

$$c_1 = a_1 + y_1$$

$$c_0, c_1 \geq 0$$

W/o borrowing constraints equivalent to standard consumer choice problem:

$$\max_{c_0, c_1} u(c_0) + \beta u(c_1)$$

$$c_0 + R^{-1} c_1 = a_0 + y_0 + R^{-1} y_1 = \text{“life-time wealth”}$$

$$c_0, c_1 \geq 0$$

- R^{-1} = price of consumption tomorrow

Optimal consumption plan

First order conditions: $\lambda =$ Lagrange multiplier on budget

$$u'(c_0) = \lambda, \qquad \beta u'(c_1) = \lambda R^{-1}$$

$$c_0 + R^{-1}c_1 = a_0 + y_0 + R^{-1}y_1$$

Euler equation: intertemporal MRS = bond price

$$\beta \frac{u'(c_1)}{u'(c_0)} = R^{-1}$$

- $\beta R = 1 \rightarrow$ smooth consumption completely, else tilt.

Power utility

$$u(c) = \frac{c^{1-1/\sigma}}{1 - 1/\sigma}$$

$$\beta \left(\frac{c_1}{c_0}\right)^{-1/\sigma} = R^{-1}, \qquad c_1 = (\beta R)^{\sigma} c_0$$

- Tilt depends on $\sigma =$ intertemporal elasticity of substitution (IES)

Response of optimal plan to increase in interest rate

$$c_0 = \left(1 + \beta^\sigma R^{\sigma-1}\right)^{-1} \left(a_0 + y_0 + R^{-1} y_1\right), \qquad c_1 = (\beta R)^\sigma c_0$$

wealth effect through $b_{-1} + y_0 + R^{-1} y_1$

- higher $R \to$ lower present value of future income
- consume less today and tomorrow

effects of change in price $1/R$:

- substitution effect
  - "consume less today because goods are cheaper tomorrow"
  - "invest today because returns are high"
- income effect
  - "consume more today because goods cheaper tomorrow"
  - "no need to invest today because returns high"

Intertemporal Elasticity of Substitution

Defined by:

$$\left.\frac{d \log \frac{c_1}{c_0}}{d \log \frac{1}{R}}\right|_{\bar{U}} = 1/\sigma.$$

IES governs relative strength of substitution effect

- log utility $\sigma = 1$: effects cancel, only get wealth effect

$\sigma > 1$: substitution effect dominates, $c_0 \downarrow$ for given wealth

$\sigma < 1$: income effect dominates, $c_0 \uparrow$ for given wealth

Optimal savings

Solve for expenditure on bonds

$$R^{-1}a_1 = a_0 + y_0 - c_0$$
$$= \frac{\beta^\sigma R^{\sigma-1}}{1+\beta^\sigma R^{\sigma-1}}(a_0 + y_0) - \frac{1}{1+\beta^\sigma R^{\sigma-1}} R^{-1} y_1$$

- 1st term: save to spread cash on hand $a_0 + y_0$ to future
- 2nd term: borrow to move forward future income y_1
- sign of a_1 depends on income profile, preferences, R

Response to increase in interest rates

- no future labor income: agent is net lender
  no wealth effect; income & subst effects: save more iff $\sigma > 1$
- no cash on hand: net borrower, borrow less (save more) for any σ
- more generally combine effects...

Borrowing constraint

Add lower bound on a_1

$$\max_{c_0, c_1, a_1} u(c_0) + \beta u(c_1)$$

$$c_0 + R^{-1} a_1 = a_0 + y_0 =: w_0$$

$$c_1 = a_1 + y_1$$

$$c_0, c_1 \geq 0, \qquad R^{-1} a_1 \geq -\phi$$

Optimal consumption plan

- add upper bound on c_0

$$c_0 \leq w_0 + \phi$$

- if unconstrained solution satisfies bound, it is still optimal
- otherwise $R^{-1} a_1 = -\phi$, $c_0 = w_0 + \phi$

The Marginal Propensity to Consume

MPC gives response to change in cash on hand

- unconstrained consumer $dc_0/dw_0 = \left(1 + \beta^\sigma R^{\sigma-1}\right)^{-1} < 1$
- constrained consumer $dc_0/dw_0 = 1$

The finite horizon problem without uncertainty

Income Profile Over the Life-Cycle

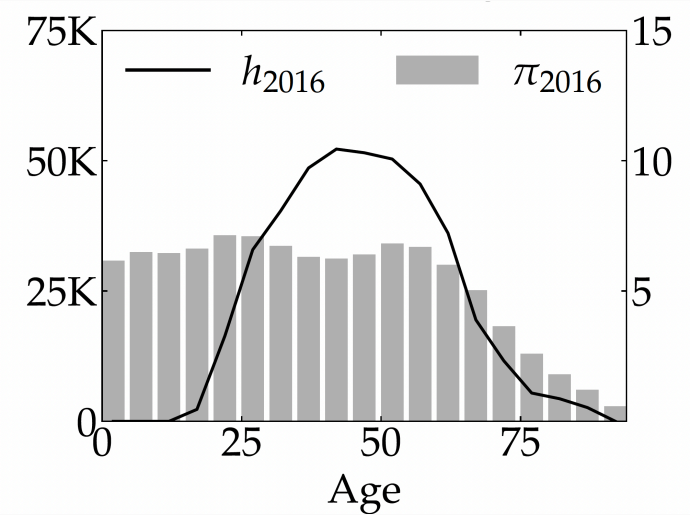

Figure 1: Source: "Demographics, Wealth, and Global Imbalances in the Twenty-First Century"

Average Asset Holding Extremely Correlated with Age

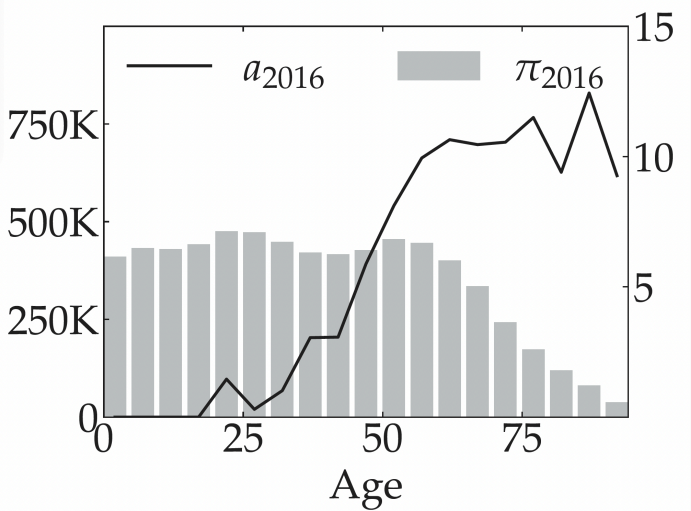

Figure 2: Source: "Demographics, Wealth, and Global Imbalances in the Twenty-First Century"

Finite horizon problem

- Each individual lives T periods
- Period utility function $u(c_t)$, discounting β
- Individuals supply inelastically one unit of labor each period
- Receive labor income we_t
  - w: wage
  - e_t: assume it's deterministic and hump-shaped over the life-cycle
- Have access to one-period risk-free bonds, a, with net interest rate r.
- Borrowing constraint $\underline{a}$

Sequential Formulation

- Sequential formulation is:

$$\max_{\{c_t, a_{t+1} \geq \underline{a}\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t u(c_t)$$

$$\text{s.t. } c_t + a_{t+1} = (1+r)a_t + e_t w \quad \forall t$$

$$a_0 \text{ given}$$

- If no borrowing constraint, analytical solution, akin to two period model
- If $\beta(1+r) = 1$ perfect consumption smoothing

Adding Risk

Models with Risk

- Event $s_t \in S$. The set $S = \{1, 2, ..., N\}$
- Example: randomly fluctuating endowments
  - $s_t = 1 \Rightarrow$ agent 1 has endowment 2
  - $s_t = 2 \Rightarrow$ agent 2 has endowment 2
- Event history $s^t = (s_0, s_1, ..., s_t) \in S^t = S \times S \times ... \times S$
- Probability of a particular event history $\pi_t(s^t) > 0$

Endowments

- Commodities indexed by time t and by event histories s^t. A consumption allocation is:

$$\left(c^1, c^2\right) = \left\{c_t^1\left(s^t\right), c_t^2\left(s^t\right)\right\}_{t=0, s^t \in S^t}^{\infty}$$

- Endowments

$$\left(e^1, e^2\right) = \left\{e_t^1\left(s^t\right), e_t^2\left(s^t\right)\right\}_{t=0, s^t \in S^t}^{\infty}$$

- For particular example

$$e_t^1\left(s^t\right) = \begin{cases} 2 & \text{if } s_t = 1 \\ 0 & \text{if } s_t = 2 \end{cases}$$

$$e_t^2\left(s^t\right) = \begin{cases} 0 & \text{if } s_t = 1 \\ 2 & \text{if } s_t = 2 \end{cases}$$

Preferences

- Preferences

$$u\left(c^i\right) = \sum_{t=0}^{\infty} \sum_{s^t \in S^t} \beta^t \pi_t\left(s^t\right) U\left(c_t^i\left(s^t\right)\right)$$

- Sometimes same preferences are described as

$$u\left(c^i\right) = E_0 \sum_{t=0}^{\infty} \beta^t U(c_t^i)$$

- This second version is less clear!

Arrow-Debreu Equilibrium

- Trade takes place at period 0, before any risk has been realized (in particular, before s_0 has been realized)
- Arrow-Debreu prices $p_t(s^t)$
- Only one budget constraint
- Market clearing has to hold date, by date, event history by event history

Arrow-Debreu Equilibrium

A (competitive) Arrow-Debreu equilibrium are prices $\left\{\hat{p}_t\left(s^t\right)\right\}_{t=0,s^t\in S^t}^{\infty}$ and allocations $\left(\left\{\hat{c}_t^i\left(s^t\right)\right\}_{t=0,s^t\in S^t}^{\infty}\right)_{i=1,2}$ such that

1. Given $\left\{\hat{p}_t\left(s^t\right)\right\}_{t=0,s^t\in S^t}^{\infty}$, for $i=1,2$, $\left\{\hat{c}_t^i\left(s^t\right)\right\}_{t=0,s^t\in S^t}^{\infty}$ solves

$$\max_{\left\{c_t^c(s^t)\right\}_{t=0,s^t\in s^t}} \sum_{t=0}^{\infty}\sum_{s^t\in S^t}\beta^t\pi_t\left(s^t\right)U\left(c_t^i\left(s^t\right)\right)$$

s.t.

$$\sum_{t=0}^{\infty}\sum_{s^t\in S^t}\hat{p}_t\left(s^t\right)c_t^i\left(s^t\right)\leq\sum_{t=0}^{\infty}\sum_{s^t\in S^t}\hat{p}_t\left(s^t\right)e_t^i\left(s^t\right)$$

$$c_t^i\left(s^t\right)\geq 0 \text{ for all } t,\ \text{all } s^t\in S^t$$

2.

$$\hat{c}_t^1\left(s^t\right)+\hat{c}_t^2\left(s^t\right)=e_t^1\left(s^t\right)+e_t^2\left(s^t\right)\ \text{ for all } t,\ \text{all } s^t\in S^t$$

Characterization of Equilibrium Prices

- First order conditions

$$\beta^t \pi_t\left(s^t\right) U'\left(c_t^i\left(s^t\right)\right) = \mu p_t\left(s^t\right)$$
$$\pi_0\left(s_0\right) U'\left(c_0^i\left(s_0\right)\right) = \mu p_0\left(s_0\right)$$

- Combining yields

$$\frac{p_t\left(s^t\right)}{p_0\left(s_0\right)} = \beta^t \frac{\pi_t\left(s^t\right)}{\pi_0\left(s_0\right)} \frac{U'\left(c_t^i\left(s^t\right)\right)}{U'\left(c_0^i\left(s_0\right)\right)}$$

- This immediately implies that

$$\frac{U'\left(c_t^1\left(s^t\right)\right)}{U'\left(c_0^1\left(s_0\right)\right)} = \frac{U'\left(c_t^2\left(s^t\right)\right)}{U'\left(c_0^2\left(s_0\right)\right)}$$

or

$$\frac{U'\left(c_t^2\left(s^t\right)\right)}{U'\left(c_t^1\left(s^t\right)\right)} = \frac{U'\left(c_0^2\left(s_0\right)\right)}{U'\left(c_0^1\left(s_0\right)\right)}$$

Characterization of Equilibrium Prices

- Ratio of marginal utilities between the two agents is constant over time and across states of the world.
- With CRRA period utility

$$\left(\frac{c_t^2\left(s^t\right)}{c_t^1\left(s^t\right)}\right)^{-\sigma} = \left(\frac{c_0^2\left(s^0\right)}{c_0^1\left(s^0\right)}\right)^{-\sigma}$$

- Thus there exist $\theta^i \geq 0$ with $\sum_i \theta^i = 1$ such that

$$c_t^i\left(s^t\right) = \theta^i e_t\left(s^t\right)$$

  with $e_t\left(s^t\right) = \sum_i e_t^i\left(s^t\right)$.
- Using this and normalizing $p_0\left(s_0\right) = 1$ we find

$$p_t\left(s^t\right) = \beta^t \frac{\pi_t\left(s^t\right)}{\pi_0\left(s_0\right)}\left(\frac{c_t^i\left(s^t\right)}{c_0^i\left(s_0\right)}\right)^{-\sigma}$$

$$\beta^t \pi_t\left(s^t\right)\left(e_t\left(s^t\right)\right)^{-\sigma}$$

Empirical Implications of Equilibrium

- Perfect consumption insurance: individual consumption of all households responds proportionally to aggregate shocks $e_t\left(s^t\right)$, but not at all to individual household endowment shocks $e_t^i\left(s^t\right)$ that leave aggregate endowments unchanged.

$$c_t^i\left(s^t\right) = \theta^i e_t\left(s^t\right)$$

- Asset prices and the representative household:

$$p_t\left(s^t\right) = \beta^t \frac{\pi_t\left(s^t\right)}{\pi_0\left(s_0\right)} \left(\frac{e_t\left(s^t\right)}{e_0\left(s_0\right)}\right)^{-\sigma}$$

- Arrow Debreu prices (and thus all other asset prices) are the same in this economy and in a representative agent economy where the representative agent has endowment stream $\left\{e_t\left(s^t\right)\right\}$ and CRRA utility with RRA σ.

Sequential Markets

- Previously: one-period IOU's
- Now: assume one period state-contingent IOU's. Complete markets!
- Let $q_t\left(s^t, s_{t+1} = j\right)$ denote price at period t of a contract that pays one unit of consumption in period $t+1$ if and only if $t+1$ event is $s_{t+1} = j$
- Let $a^i_{t+1}\left(s^t, s_{t+1}\right)$ be the quantities Arrow securities bought at period t by agent i.
- Budget constraint of agent i

$$c^i_t\left(s^t\right) + \sum_{s_{t+1} \in S} q_t\left(s^t, s_{t+1}\right) a^i_{t+1}\left(s^t, s_{t+1}\right) \leq e^i_t\left(s^t\right) + a^i_t\left(s^t\right) \quad \forall t, s^t$$

- Can prove equivalence between Arrow Debreu and Sequential Markets Equilibrium

Numerical solution

Finite Horizon Sequential Formulation Again

- Let's go back to the finite horizon problem without risk:

$$\max_{\{c_t, a_{t+1} \geq \underline{a}\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t u(c_t)$$
$$\text{s.t. } c_t + a_{t+1} = (1+r)a_t + e_t w \quad \forall t$$
$$a_0 \text{ given}$$

- Assume we want to solve numerically, how?

Recursive Formulation

- Recursive formulation is:

$$V(t,a) = \max_{\{c,a' \geq \underline{a}\}} \quad u(c) + \beta V(t+1, a')$$
$$\text{s.t. } c + a' = (1+r)a + e_t w$$
$$V(T+1, a) = 0$$

- Age and assets are the state variables
- Productivity not a state variable here because age is sufficient
- No bequest ($V(T+1, a) = 0$)

Final Condition

- At time T continuation value is 0
- $\rightarrow$ So it is optimal to save 0 assets
- Consumption at the end of life is:

$$c(T, a) = (1 + r)a + e_T w$$

- And final V:

$$V(T, a) = u\left((1 + r)a + e_T w\right)$$

Computation

- Start with $V(T, a) = u\left((1+r)a + e_T w\right)$
- for $t = T-1, \ldots, 0$, use $V(t+1, a)$ to solve

$$V(t, a) = \max_{\{c, a' \geq \underline{a}\}} \quad u(c) + \beta V\left(t+1, a'\right)$$

$$\text{s.t. } c + a' = (1+r)a + e_t w$$

- We have to discuss discretization and maximization.

Discretization and grid generation

- Since we have a continuous state space, we need to discretize it into a grid
- Huge literature on numerical analysis on how to efficiently generate grids.

Uniform Grid

- Decide lower bound and upper bound
- Decide how many points in the grid
  - Trade-off between speed and precision
- Distribute them uniformly in the state space
- Problem: assets are unbounded $\rightarrow$ how do we choose upper bound?
  - We want to be able to characterize the very wealthy
  - Choose an upper bound value. Check that it is not binding (e.g. savings function crosses 45 degrees line). Otherwise, relax it

Non-Uniform Grid

- Use economic theory or error analysis to evaluate where to accumulate points.
- Standard argument: close to curvatures of the value function or kinks.
- Do not need many points where value function is close to linear
- In this model, more points close to the borrowing constraint. Why?
  - Can use exponential or double-exponential grid
- More generally, problem: this is a heuristic argument.
- For more advanced methods, including adaptive sparse grids: `https://github.com/SparseGridsForDynamicEcon/SparseGrids_in_econ_handbook`

Maximization

- We need to apply the max operator.
- Costly step of value function iteration.
- **Brute force (always works): check all the possible choices in the grid.**
  - Can be sped up by using monotonicity of choices (more on this later)
- Using a Newton or quasi-Newton algorithm.
- Endogenous grid method skips this step entirely

Endogenous Grid Method

- Proposed by Carroll (2005).
- When constraint not binding, policy function satisfies first-order condition:

$$u'(c) = \beta V_a(a')$$

so:

$$c = (u')^{-1}(\beta V_a(a'))$$

- Idea of endogenous grid method is to start with V_a, defined on end of period assets, and back out consumption on *beginning of period assets*.
- Constraint can be imposed ex-post

Endogenous Grid Method Algorithm I

- The algorithm is as follows:

1. Start with $V_{a,T}(a') = (1+r)u'(c') = (1+r)u'((1+r)a' + e_T w)$
   - Notice $V_{a,T}$ is defined on a grid a', or on end-of-period assets
2. Get optimal consumption $c_{T-1}^{endog} = (u')^{-1}(\beta V_{a,T})$
3. From the budget constraint, we can find a^* such that:

$$c_{T-1}^{endog} + a'_{T-1} = \underbrace{(1+r)a^*}_{\text{endogenous resources } m} + we_{T-1}$$

   this means we have found policy functions $c_{T-1}^{endog}(a^*)$ and $a'_{T-1}(a^*)$.
4. However, a^* does not necessarily correspond to our grid a. To find $a'_{T-1}(a)$ we can interpolate

$$(\underbrace{c_{T-1}^{endog} + a'_{T-1} - we_{T-1}}_{m}, \underbrace{a'_{T-1}}_{f(m)}) \text{ on } (1+r)a$$

Endogenous Grid Method Algorithm II

- Continued:

5. To impose the borrowing constraints, set any $a' < 0$ to 0.
6. To get $c_{T-1}(a)$, simply solve $c = (1+r)a + we_{T-1} - a'$
   - In more complicated cases, it may be worth to calculate in advance consumption when borrowing constraints are binding instead of doing this every iteration.
7. To continue the recursion, use the envelope condition

$$V_{a,T-1} = (1+r)u'(c_{T-1})$$

EGM code

```python
# Initialize policy functions
c = np.zeros((T, len(a_grid)))  # Consumption policy
a = np.zeros((T, len(a_grid)))  # Savings policy

# Final period: consume all resources
a[-1, :] = 0  # No savings nor borrowing in the last period
c[-1, :] = y_profile[-1] + (1 + r) * a_grid  # Consumption in the final period
c[-1, c[-1,:]<0] = 0 # Consumption cannot be negative

for tau in range(T-2, -1, -1):  # Loop from period T-1 to 0

    # Step 1: Forming Expectations
    c_prime = c[tau+1, :]  # Consumption policy in the next period
    Va = beta*(1 + r) * c_prime ** (-1 / eis)

    # Step 2: Finding endogenous consumption and resources
    c_endo =  Va ** (-eis)  # FOC
    m_endo = c_endo + a_grid - y_profile[tau]  # endogenous resources

    # breakpoint()

    # Step 3: Evaluating in our grids
    a[tau, :] = np.interp((1+r)*a_grid, m_endo, a_grid)  # Current savings

    # Step 4: constraints (a>amin, c>0)
    a[tau, :] = np.maximum(a[tau, :], a_grid[0])
    c[tau, :] = y_profile[tau] + (1 + r) * a_grid - a[tau, :]
    c[tau, c[tau,:]<0] = 0 # Consumption cannot be negative

return a, c
```

- Sample code for endogenous grid method in a life cycle model

When is EGM inconvenient?

- When it is not possible to use V_a to find c analytically
  - For example: Non-separable preferences between consumption and leisure
  - Certain types of non-homothetic preferences
- When there are two or more endogenous state variables
  - Method requires interpolating over irregular grids, which is hard in two or more dimensions
  - Delaunay interpolation can help
- In general more complex problems
- What to do? Use brute force or use other methods to speed up.